Practical no. 8B

Aim: Create a simple Web API that returns Student data in JSON format.

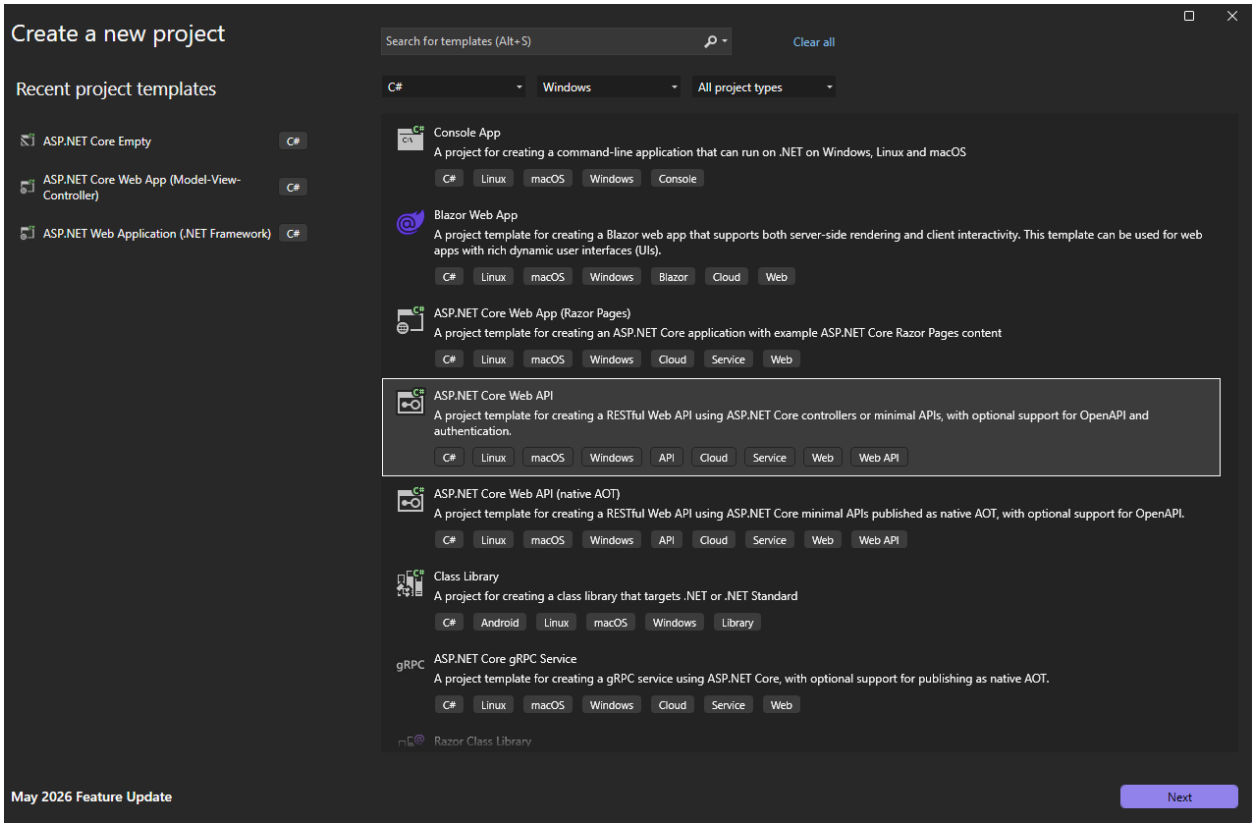
Software Required:

Visual Studio 2026

Procedure:

STEP 1: CREATE NEW PROJECT

1. Open Visual Studio 2026.
2. Click: Create a New Project
3. Select: **ASP.NET Core Web API**
4. Click Next.
5. Enter: Project Name: Practical 8B
6. Click Next.
7. Select: Framework: .NET 10 (or latest)
   Authentication Type: None
   Check: Configure for HTTPS
8. Click Create.



Step 2: Create Student Model

1. Click View.
2. Click Solution Explorer..
3. Right click project name.
4. Add→ New folder.
5. Enter Name: **Models**
6. Right Click Models.
7. Click Add.
8. Click New item.
9. Select Class.
10. Give Name **Student.cs.**
11. Click Add.

Student.cs

```
namespace Practical_8b.Models
{
   public class Student
   {
      public int Id { get; set; }

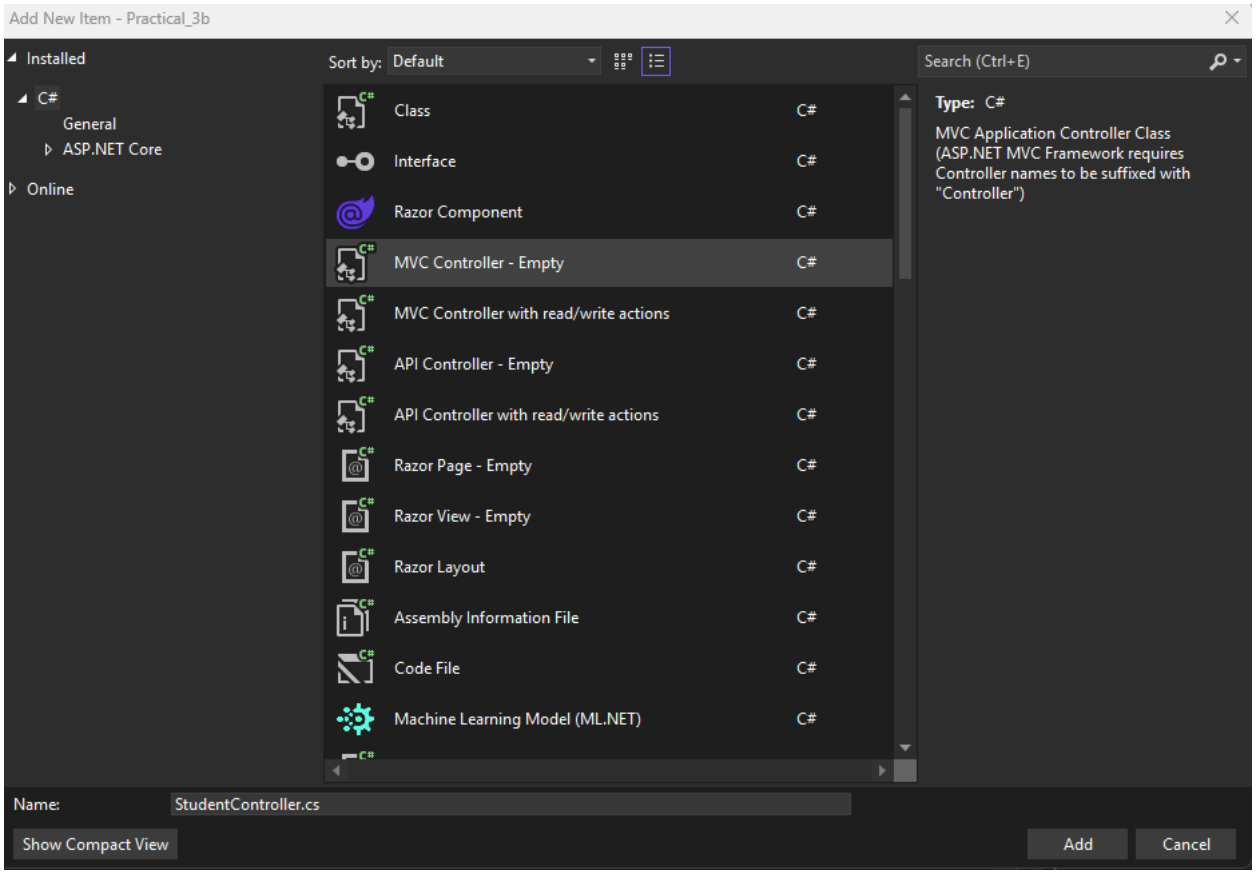
      public string Name { get; set; }

      public string Course { get; set; }
   }
}
```

Step 3: Create Controller

Create a StudentController inside the Controllers folder.

1. Right Click Controllers.
2. Click Add.
3. Click New item.
4. Select MVC Controller - Empty.
5. Give Name **StudentController**.
6. Click Add.



StudentController.cs

```
using Microsoft.AspNetCore.Mvc;
using Practical_3b.Models;

namespace Practical_8b.Controllers
{
   [ApiController]
   [Route("api/[controller]")]
   public class StudentController : ControllerBase
   {
      [HttpGet]
      public IActionResult GetStudents()
      {
         List<Student> students = new List<Student>()
         {
            new Student
            {
```

```
                Id = 101,
                Name = "Rahul",
                Course = "BSc IT"
            },

            new Student
            {
                Id = 102,
                Name = "Priya",
                Course = "BSc CS"
            },

            new Student
            {
                Id = 103,
                Name = "Amit",
                Course = "BCA"
            }
        };

        return Ok(students);
    }
  }
}
```

Step 4: Modify Program.cs

Solution Explorer → Program.cs

```
var builder = WebApplication.CreateBuilder(args);

// Add Controller Services
builder.Services.AddControllers();

var app = builder.Build();

// Configure HTTP Request Pipeline
app.UseHttpsRedirection();

// Map Controllers
app.MapControllers();
```

app.Run();

Step 8: Build the Project

Click: Build → Build Solution

OR

Ctrl + Shift + B

Step 9: Run the Application

Ctrl + F5

Step 10: Test API

Manually add: /api/student

Example:

https://localhost:44315/api/student

